SQL Fundamentals: SELECT & Filtering

EXECISE 1

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--EXERCISE 1

--01: SELECT & FILTERING
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--Step 1: Create a database/catalog
--CREATE CATALOG IF NOT EXISTS workspace;

--Step 2: Create a schema
--CREATE SCHEMA IF NOT EXISTS workspace.salary;
```

```
--Question 1: Retrieve all columns from employees table
SELECT*
FROM workspace.salary.employees;
```

	id	first_name	last_name	department	salary	hire_date	city
1	1	Alice	Green	IT	70000	2009	Johannesburg
2	2	Brian	Lee	HR	45000	1985	Cape Town
3	3	Cathy	Zulu	Finance	65000	1993	Durban
4	4	David	Mokoena	Marketing	50000	2005	Pritoria
5	5	Eva	Naidoo	IT	72000	1978	Johannesburg

```
--Question 2: Find all unique departments
SELECT DISTINCT department
FROM workspace.salary.employees
```

	department
1	IT
2	HR
3	Finance
4	Marketing

```sql
--Question 3: Retrieve first and last names ordered by salary descending
SELECT first_name,
last_name
FROM workspace.salary.employees
ORDER BY salary DESC;
```

Table

	first_name	last_name
1	Eva	Naidoo
2	Alice	Green
3	Cathy	Zulu
4	David	Mokoena
5	Brian	Lee

```sql
--Question 4: Retrieve the top 3 highest-paid employees
SELECT id, first_name,
last_name,
salary
FROM workspace.salary.employees
ORDER BY salary DESC
LIMIT 3;
```

Table

	id	first_name	last_name	salary
1	5	Eva	Naidoo	72000
2	1	Alice	Green	70000
3	3	Cathy	Zulu	65000

```sql
--Question 5: Find employees in the IT department
SELECT id,
first_name,
last_name,
department
FROM workspace.salary.employees
WHERE department ='IT';
```

Table

	id	first_name	last_name	department
1	1	Alice	Green	IT
2	5	Eva	Naidoo	IT

```sql
--Question 6: Find employees in finance with salary > 60000
SELECT id,
first_name,
last_name,
department,
salary
FROM workspace.salary.employees
WHERE department = 'Finance' AND salary > 60000;
```

Table

	id	first_name	last_name	department	salary
1	3	Cathy	Zulu	Finance	65000

```sql
--Question 7: Find employees in HR or Marketing
SELECT id,
first_name,
last_name,
department
FROM workspace.salary.employees
WHERE department = 'HR' OR department = 'Marketing';
```

Table

	id	first_name	last_name	department
1	2	Brian	Lee	HR
2	4	David	Mokoena	Marketing

```sql
--Question 8: Find employees not in IT
SELECT id,
first_name,
last_name,
department
FROM workspace.salary.employees
WHERE department != 'IT';
```

Table

	id	first_name	last_name	department
1	2	Brian	Lee	HR
2	3	Cathy	Zulu	Finance
3	4	David	Mokoena	Marketing

```sql
--Question 9: Find employees in IT, HR, or Finance using IN
SELECT id,
first_name,
last_name,
department
FROM workspace.salary.employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Table

	id	first_name	last_name	department
1	1	Alice	Green	IT
2	2	Brian	Lee	HR
3	3	Cathy	Zulu	Finance
4	5	Eva	Naidoo	IT

```sql
--Question 10: Find employees in IT with salary > 65000 and city is Johannesburg
SELECT id,
first_name,
last_name,
department,
salary,
city
FROM workspace.salary.employees
WHERE department IN ('IT') AND salary > 65000 AND city = 'Johannesburg';
```

Table

	id	first_name	last_name	department	salary	city
1	1	Alice	Green	IT	70000	Johannesburg
2	5	Eva	Naidoo	IT	72000	Johannesburg